

Problem 1

- (a) Let $\sigma \in S_n$, write $\sigma = a_1 b_1 \dots a_m b_m$ where a_i, b_i are transpositions

Consider $a_k b_k$

$$a_k b_k = (ab)(cd) \text{ or } a_k b_k = (ab)(ac) \text{ s.t. } a, b, c, d \in \{1, 2, \dots, n\}$$

$$(ab)(cd) = (acb)(acd) \text{ or } (ab)(ac) = (acb)$$

Thus since we have an even amount of transpositions we can write them all as 3-cycles and thus A_n is generated by 3 cycles.

- (b) Let $(123) \in A_n$ to show all 3-cycles conjugate we only need to show they conjugate with (123) since conjugation is an equivalence class

Consider (abc) and $(1a)(2b)(3c)$ for $a, b, c \in \{1, 2, \dots, n\}$ if $(1a)(2b)(3c)$ is even then

$$(1a)(2b)(3c)(abc)((1a)(2b)(3c))^{-1} = (123)$$

If it is not even then consider $(1a)(2b)(3c)(45)$

$$(1a)(2b)(3c)(45)(abc)((1a)(2b)(3c)(45))^{-1} = (123)$$

Thus since they are all conjugate to (123) we know all 3-cycles are thus conjugate to one another.

- (c) i. Let $N \trianglelefteq A_n$ if $N = \{e\}$ then done so assume $N \neq \{e\}$

Let $e \neq \sigma \in N$ s.t. it has the most fixed points.

- ii. Let $\sigma = (a_1 b_1)(a_2 b_2) \dots (a_t b_t)$ bet disjoint transpositions.

t is not 1 since this would mean σ is odd so $t \geq 2$

Since these are distinct this means we fix $n - 2t \geq n - 4$ elements, and since $n \geq 5 \implies$

$n - 2t \geq n - 4 \geq 1$ thus there is at least one element that is fixed by σ

Let $r \in \{1, \dots, n\}$ s.t. r is fixed in σ

Let $\tau = (a_1 b_1 r)$

Consider $\tau \sigma \tau^{-1} \sigma^{-1}$

Since N is normal we know $\tau \sigma \tau^{-1} \in N$ and since $\sigma^{-1} \in N$ we know $[\tau, \sigma] \in N$

$$= \tau(a_1 b_1) \tau^{-1} \tau(a_2 b_2) \tau^{-1} \dots \tau(a_t b_t) \tau^{-1} \sigma$$

$$= (b_1 r)(a_2 b_2) \dots (a_t b_t) \sigma$$

$$= (b_1 r)(a_1 b_1) = (a_1 r b_1)$$

Thus $[\tau, \sigma]$ fixes $n - 3$ points

$n - 2t \leq n - 4 < n - 3$ thus the commutator fixes more points and is not the identity, thus σ cannot be written as disjoint transpositions.

- iii. Since 3-cycles cant be written as two disjoint transpositions if σ is a 3 cycle then we are done so assume not

The only way for a permutation to move an even amount of elements if it can be written as disjoint transpositions thus it σ cannot move only 4 elements then it could be written

as 2 disjoint transpositions thus σ must move at least 5 elements.

iv. Thus $\sigma = (a_1 a_2 a_3 \dots)(\dots) \dots (\dots)$

Let r, s be two elements not in $\{a_1, a_2, a_3\}$ that are moved by σ .

Let $\tau = (a_3 r s)$

Consider $\tau \sigma \tau^{-1} \sigma^{-1}$

Since N is normal we know $\tau \sigma \tau^{-1} \in N$ and since $\sigma^{-1} \in N$ we know $[\tau, \sigma] \in N$

Let x be fixed by σ then $\tau \sigma \tau^{-1} \sigma^{-1}(x) = \tau \sigma \tau^{-1}(x)$ and since τ only touches things that are moved by σ thus $= \tau \sigma(x) = \tau(x) = x$

Thus if x is fixed by σ it is fixed by $[\tau, \sigma]$

Consider a_2 , σ moves a_2 now consider $\tau \sigma \tau^{-1} \sigma^{-1}(a_2) = \tau \sigma \tau^{-1}(a_1) = \tau \sigma(a_1) = \tau(a_2) = a_2$

Thus the commutator fixes a_2 thus this is a contradiction thus σ must be a 3-cycle

Thus we know N has a 3-cycle and by proven above since N is normal it is closed under conjugation thus we have all 3-cycles in N and since 3-cycles generate A_n thus $N = A_n$

Problem 2

$|S_p| = p!$ thus the order of a p -Sylow is just p

Thus the number of distinct p cycles is $(p-1)!$ thus there are $(p-1)!$ groups generated by each of these p cycles

However, proven in last homework since in these groups are of order p we know it has $p-1$ generators thus we are overcounting by a factor of $p-1$ since we count each group $p-1$ times thus

$$n_p = \frac{(p-1)!}{p-1} = (p-2)!$$

Thus by Sylow 3 we know $n_p = (p-2)! \equiv 1 \pmod{p} \implies (p-1)! \equiv (p-1) \pmod{p} \implies (p-1)! \equiv -1 \pmod{p}$

Problem 3

Consider $SL_2(F_5) = \{2 \times 2 \text{ matrices} \mid \det = 1\}$

Thus for matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $SL_2(F_5)$ we know $ad - bc = 1$

Consider the first column we have $5^2 - 1$ choices since all it has to be is non zero

Then for the second column we have 5 choices for b but then d is fixed to satisfy $ad - bc = 1$ thus we have 120 choices thus $|SL_2(F_5)| = 120$

Thus since $PSL_2(F_5) = \frac{SL_2(F_5)}{\{\pm I\}}$ we know $|PSL_2(F_5)| = \frac{|SL_2(F_5)|}{|\{\pm I\}|} = 60$

Let x and y be two classes of matrices

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

Consider A

$A^5 = I$ thus $\langle AI \rangle = \langle A \rangle$ thus $\langle A \rangle$ is a subgroup of order 5

Consider B

$B^5 = I$ thus $\langle BI \rangle = \langle B \rangle$ thus $\langle B \rangle$ is a subgroup of order 5

Since $A \neq B$ and $-A \neq B$ we know that these are distinct in $PSL_2(F_5)$

Thus since a group of order 60 has more than 1 subgroup of order 5 we know it must have 5 by Sylow 3, and thus it must be isomorphic to A_5

Problem 4

- (a) Consider $SL_2(F_7) = \{2 \times 2 \text{ matrices} \mid \det = 1\}$

Thus for matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $SL_2(F_7)$ we know $ad - bc = 1$

Consider the first column we have $7^2 - 1$ choices since all it has to be is non zero

Then for the second column we have 7 choices for b but then d is fixed to satisfy $ad - bc = 1$ thus we have 120 choices thus $|SL_2(F_7)| = 7(7^2 - 1) = 336$

Thus since $PSL_2(F_7) = \frac{SL_2(F_7)}{\{\pm I\}}$ we know $|PSL_2(F_7)| = \frac{|SL_2(F_7)|}{|\{\pm I\}|} = 168$

- (b) Let x and y be two classes of matrices

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Consider A

$A^7 = I$ thus $\langle AI \rangle = \langle A \rangle$ thus $\langle A \rangle$ is a subgroup of order 7

Consider B

$B^7 = I$ thus $\langle BI \rangle = \langle B \rangle$ thus $\langle B \rangle$ is a subgroup of order 7

Since $A \neq B$ and $-A \neq B$ we know that these are distinct in $PSL_2(F_7)$

Thus since a group of order 168 has more than 1 subgroup of order 7 we know it must have 8 by Sylow 3.

- (c) Let $N \trianglelefteq G$ s.t. $[G : N] = 2$ this means that $|N| = 84$ thus $n_7 = 1$ by Sylow 3. However, this is a contradiction since proven in class if $P \trianglelefteq N \trianglelefteq G$ and P is a p -Sylow then $P \trianglelefteq G$ this implies that the one normal seven Sylow is normal in G , but this is a contradiction of Sylow 2 thus it can't be index 2.

- (d) Suppose $[G : N] = 3$ this means $|N| = 56$ thus $n_7 = 1, 8$ by Sylow 3. It must have 8 by shown above thus it has all 7-Sylow subgroups of G . Thus, it must contain both H and K

Thus by group closure it contains element $u = xy^2x$ s.t. $H = \langle x \rangle$ and $K = \langle y \rangle$

$$u = AB^2A$$

$$u^2 = xy^2xxy^2x = xy^2x^2y^2x$$

$$u^3 = xy^2x^2y^2xxy^2x = xy^2x^2y^2x^2y^2x = AB^2A^2B^2A^2B^2A = I$$

This is a contradiction since N is a group of order 56 and 3 does not divide 56 thus it cannot have a subgroup of order 3 thus $[G : N] \neq 3$

- (e) Assume $[G : N] \neq 6$ thus $|N| = 28$ thus by Sylow 3 $n_7 = 1$ this is a contradiction by similar proof above thus $[G : N] \neq 6$

- (f) Assume G is not simple and let $N \trianglelefteq G$ be a maximal proper normal subgroup this means that G/N must be simple thus we only have to check for groups of prime order that divide 168 and since 60 does not divide 168 we don't have to check this

Thus $[G : N] = 7$ since we checked for 2,3 thus $|N| = 24$

- (g) Thus by Sylow 3 we know that $n_2(N) = 1, 3$

Assume $n_2(N) = 1$ then we know there is a normal 2-Sylow H of N that is normal in G . Thus, G/H is a subgroup of order 21 since $168/8 = 21$ thus by Sylow 3 this has a normal 7-Sylow $\bar{K}$. Thus, by proven in homework we know corresponds to a normal subgroup in G called K of G thus since $|K/H| = \bar{K}$ we know that $|K| = 8 \times 7 = 56$

Thus $[G : K] = 3$ which is a contradiction thus $n_2(N) = 3$

- (h) Thus G acts on the 3 groups 2-Sylow groups by conjugation thus there exists a nontrivial homomorphism $\varphi : G \rightarrow S_3$. This map cannot be injective since $|G| > |S_3|$ thus the kernel is nontrivial, thus $\ker(\varphi) = K$ is a proper normal subgroup of G .

By the first isomorphism theorem we know that $G/K \cong \text{Im}(\varphi) = I$ and the $|I| \mid 6$ and is not 1 thus we get $|I| = 3, 6$ thus $[G : K] = |I| = 3, 6$ and above we have shown these are contradictions thus G must be simple

Problem 5

- (a) Look at last homework the order of $GL_2(F_2)$ is $2^3(2-1)(2^2-1)(2^3-1) = 168$
- (b) Since $x^7 = y^7 = e$ these both form distinct 7-Sylow subgroups of G thus we know it has 8 7-Sylows, and following the same exact proof as above we can show for $N \trianglelefteq G$ we know that $[G : N] \neq 2$
 Now for $[G : N] = 3$ we know that $u = yx$ is order 3 thus using the same argument we get that $[G : N] \neq 3$. Then for index 6 we do the same argument. Then following the same steps as the proof in the last problem we can say that G is simple
- (c) Consider the subgroup of upper triangular matrices by last homework we know the order of this subgroup is order 8. Thus, this is a 2-Sylow subgroup.

Consider:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A^4 = I, \text{ and } B^2 = I, \text{ and } BAB = A^{-1}$$

Thus similar proof to homework 1 we can show that this 2-Sylow is isomorphic to D_4 and since all 2-Sylows are conjugate we know that they are all isomorphic to each other. Thus the isomorphism class for the 2-Sylows is just D_4

Problem 6

- (a) Let
- $\sigma : (X, Y, Z) \rightarrow (Y, Z, X)$

Since it is a 3 cycle is order 3, and it preserves the Klein quartic equation.

$$Y^3Z + Z^3X + X^3Y = 0$$

- (b) Let
- $\tau : (X, Y, Z) \rightarrow (\omega X, \omega^4 Y, \omega^2 Z)$
- s.t.
- $\omega = e^{2\pi i/7}$

$\tau^7 = (\omega^7 X, \omega^{28} Y, \omega^{14} Z) = (X, Y, Z)$ This is order 7, and it preserves the Klein quartic equation.

$$(\omega X)^3(\omega^4 Y) + (\omega^4 Y)^3(\omega^2 Z) + (\omega^2 Z)^3(\omega X)$$

$$\omega^7 X^3 Y + \omega^{14} Y^3 Z + \omega^7 Z^3 X = 0$$

- (c) Let
- $A = \langle \sigma, \tau \rangle$
- . Consider
- $\sigma\tau\sigma^{-1}(X, Y, Z) = (\omega^4 X, \omega^2 Y, \omega Z) = \tau^4(X, Y, Z) = \tau^{-3}(X, Y, Z)$

Thus $A = \langle \sigma, \tau \mid \sigma^3 = \tau^7 = e, \sigma\tau\sigma^{-1} = \tau^{-3} \rangle$

Since we know that $\sigma\tau\sigma \in \langle \tau \rangle$ we know that $\langle \tau \rangle \trianglelefteq A$ since we only have to check generators.

Since $\sigma\tau \neq \tau\sigma$ we know A is not abelian, and since A is generated by σ, τ we know that for all $a \in A$ $a = \tau^i \sigma^j$ or $\sigma^j \tau^i$ but since $\langle \tau \rangle$ is normal we know that $\sigma^j \tau^i = \tau^k \sigma^j$ thus all elements in A can be written of the form $\tau^i \sigma^j$, and since $\langle \tau \rangle$ only has order 1,7 and $\langle \sigma \rangle$ only has 1,3 we know that $\langle \tau \rangle \cap \langle \sigma \rangle = \{e\}$ thus by previous homework we know that $|A| = |\langle \tau \rangle| |\langle \sigma \rangle| / 1 = 21$ thus since $21 = 3 \times 7$ proven in homework since $3 \mid (7 - 1)$ we know $A \cong C_7 \rtimes C_3$